

Student ID

Name

Problem

ADA Demo Exam — Answer Sheet

Write in the three boxes at the top of EVERY page:

left: your student ID (e.g. B11902001)

middle: your name

right: the problem number, prefixed (Q1, Q2, Q3, Q4)

Use one page per problem. Each problem is worth 10 points.

Q1. Binary search complexity (10 points)

State the worst-case running time of binary search on a sorted array of n elements and prove your bound by analysing the recurrence $T(n) = T(n/2) + O(1)$.

Q2. Greedy interval scheduling (10 points)

You are given n intervals. Prove that repeatedly choosing the compatible interval with the earliest finish time yields a maximum-size set of non-overlapping intervals (exchange argument expected).

Q3. Longest increasing subsequence (10 points)

Give a dynamic-programming algorithm for the longest increasing subsequence of a sequence of n numbers. Define your subproblems, give the recurrence, and state the running time. An $O(n \log n)$ refinement earns full marks.

Q4. BFS shortest paths (10 points)

Prove or disprove: in an unweighted graph, breadth-first search from s computes a shortest path from s to every reachable vertex. Justify with an invariant over BFS layers.